

## CONCEPCION, Paraguay

WMO: 861340

Lat: 23.442S

Lon: 57.427W

Elev: 253

StdP: 14.56

Time Zone: -4.00 (PRY)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 6             | 43.1       | 46.9 | 34.7                          | 29.7 | 52.8 | 38.2 | 34.1 | 53.2 | 23.3                  | 79.4 | 21.0 | 78.3 | 2.7                   | 180  | 0.360 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 17.9                   | 100.1           | 75.0 | 98.2 | 75.4 | 96.5 | 75.6 | 80.8                | 90.8 | 80.0 | 90.1 | 79.5 | 89.8 | 12.8                 | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.2                            | 147.8 | 85.7 | 77.3 | 143.5 | 85.1 | 76.5 | 139.6 | 84.5 | 44.6          | 90.7 | 43.8 | 90.0 | 43.3 | 92.0 | 87.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 21.5              | 18.8 | 17.3 | DB                         | 36.4 | 103.4              | 4.1 | 2.1                                                | 33.4 | 104.9      | 31.0 | 106.2      | 28.7 | 107.3      | 25.7 | 108.9 |
|                   |      |      | WB                         | 35.4 | 82.2               | 3.9 | 0.6                                                | 32.6 | 82.6       | 30.3 | 83.0       | 28.1 | 83.3       | 25.3 | 83.8  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 75.9   | 83.2  | 82.5  | 80.6  | 76.3  | 69.7  | 67.3  | 67.6  | 70.9  | 74.0  | 77.9  | 79.1  | 82.3  |
|                                                                        | DBStd       | 9.12   | 3.37  | 4.15  | 4.64  | 6.55  | 7.15  | 9.08  | 9.63  | 10.20 | 9.53  | 6.97  | 5.45  | 4.55  |
|                                                                        | HDD50       | 6      | 0     | 0     | 0     | 0     | 0     | 2     | 3     | 1     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 319    | 0     | 0     | 0     | 9     | 36    | 86    | 94    | 63    | 29    | 2     | 0     | 0     |
|                                                                        | CDD50       | 9466   | 1029  | 909   | 948   | 790   | 611   | 522   | 547   | 649   | 721   | 866   | 872   | 1002  |
|                                                                        | CDD65       | 4303   | 564   | 489   | 483   | 348   | 182   | 156   | 173   | 246   | 300   | 403   | 422   | 537   |
|                                                                        | CDH74       | 46058  | 6593  | 5461  | 5050  | 3177  | 1378  | 1147  | 1518  | 2757  | 3577  | 4485  | 4675  | 6240  |
|                                                                        | CDH80       | 21820  | 3202  | 2612  | 2406  | 1373  | 459   | 368   | 596   | 1352  | 1881  | 2269  | 2261  | 3041  |
| (14) Wind                                                              | WSAvg       | 7.0    | 6.1   | 6.1   | 5.5   | 5.9   | 6.1   | 6.8   | 7.7   | 8.5   | 8.7   | 8.3   | 7.3   | 7.0   |
| (15) Precipitation                                                     | PrecAvg     | 56.0   | 6.6   | 5.9   | 5.5   | 5.4   | 4.6   | 2.8   | 1.5   | 1.1   | 3.0   | 5.7   | 7.7   | 6.2   |
|                                                                        | PrecMax     | 84.1   | 16.8  | 18.8  | 11.9  | 13.1  | 9.1   | 10.9  | 5.2   | 4.3   | 8.7   | 10.2  | 18.8  | 15.2  |
|                                                                        | PrecMin     | 36.4   | 1.7   | 0.9   | 0.8   | 0.1   | 0.8   | 0.1   | 0.0   | 0.2   | 0.9   | 2.4   | 1.6   | 1.6   |
|                                                                        | PrecStd     | 10.5   | 4.1   | 3.5   | 2.7   | 3.6   | 2.5   | 2.5   | 1.2   | 1.1   | 2.1   | 2.3   | 4.3   | 3.3   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 100.6 | 100.5 | 99.9  | 96.3  | 91.8  | 90.1  | 92.2  | 98.6  | 101.6 | 103.2 | 100.9 | 99.9  |
|                                                                        |             | MCWB   | 75.8  | 77.1  | 76.0  | 77.6  | 74.5  | 72.2  | 69.6  | 70.0  | 72.7  | 74.9  | 76.8  | 76.6  |
|                                                                        | 2%          | DB     | 97.4  | 97.2  | 96.9  | 93.9  | 88.9  | 87.1  | 89.2  | 95.3  | 99.0  | 98.3  | 97.5  | 97.6  |
|                                                                        |             | MCWB   | 76.6  | 76.4  | 76.0  | 77.0  | 74.3  | 71.7  | 69.6  | 70.1  | 72.4  | 75.5  | 76.3  | 77.3  |
|                                                                        | 5%          | DB     | 95.2  | 94.9  | 94.4  | 91.5  | 86.0  | 84.7  | 86.5  | 92.2  | 95.8  | 95.4  | 94.7  | 95.4  |
|                                                                        |             | MCWB   | 76.8  | 77.3  | 76.6  | 76.5  | 73.6  | 71.5  | 69.4  | 69.3  | 72.0  | 75.3  | 76.7  | 77.6  |
|                                                                        | 10%         | DB     | 93.0  | 92.2  | 91.5  | 88.7  | 82.5  | 81.7  | 83.4  | 88.6  | 91.7  | 92.1  | 91.6  | 93.1  |
|                                                                        |             | MCWB   | 77.3  | 77.4  | 76.6  | 75.8  | 72.4  | 70.6  | 68.8  | 69.0  | 71.2  | 75.0  | 75.7  | 77.9  |
|                                                                        | 0.4%        | WB     | 81.5  | 81.7  | 80.9  | 80.1  | 78.0  | 76.5  | 74.7  | 75.3  | 78.0  | 80.0  | 81.3  | 81.5  |
|                                                                        |             | MCDB   | 90.8  | 91.3  | 90.4  | 90.9  | 86.5  | 84.4  | 84.8  | 88.0  | 91.7  | 93.2  | 92.4  | 91.7  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 80.3  | 80.2  | 79.6  | 79.1  | 76.4  | 74.4  | 73.0  | 73.3  | 75.8  | 78.8  | 79.5  | 80.1  |
|                                                                        |             | MCDB   | 89.5  | 90.0  | 89.9  | 89.4  | 84.6  | 82.7  | 84.0  | 87.0  | 91.1  | 91.8  | 91.2  | 90.6  |
|                                                                        | 5%          | WB     | 79.7  | 79.5  | 78.5  | 77.9  | 74.9  | 72.6  | 71.3  | 71.7  | 74.1  | 77.5  | 78.2  | 79.6  |
|                                                                        |             | MCDB   | 89.1  | 89.3  | 89.1  | 87.8  | 82.6  | 80.9  | 82.1  | 86.1  | 89.3  | 89.8  | 89.8  | 90.1  |
|                                                                        | 10%         | WB     | 78.9  | 78.5  | 77.7  | 76.8  | 73.3  | 71.1  | 69.6  | 70.2  | 72.5  | 76.0  | 77.2  | 78.8  |
|                                                                        |             | MCDB   | 88.5  | 88.1  | 88.2  | 86.1  | 80.8  | 79.4  | 80.3  | 84.8  | 87.0  | 88.2  | 88.3  | 89.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 17.9  | 17.4  | 18.6  | 17.6  | 16.5  | 17.1  | 19.8  | 22.9  | 22.7  | 19.8  | 19.6  | 17.5  |
|                                                                        |             | MCDBR  | 21.9  | 21.5  | 23.1  | 21.3  | 20.9  | 20.6  | 23.9  | 26.6  | 27.6  | 24.4  | 23.8  | 21.4  |
|                                                                        |             | MCWBR  | 6.5   | 6.3   | 7.1   | 8.0   | 9.3   | 9.3   | 9.9   | 9.3   | 9.4   | 8.4   | 8.1   | 6.8   |
|                                                                        | 5% WB       | MCDBR  | 16.9  | 17.0  | 18.5  | 18.5  | 17.4  | 17.6  | 20.7  | 23.1  | 23.0  | 20.7  | 18.8  | 17.3  |
|                                                                        |             | MCWBR  | 6.8   | 6.9   | 7.4   | 8.2   | 9.5   | 9.6   | 10.9  | 10.9  | 9.5   | 8.9   | 7.9   | 7.1   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.416 | 0.414 | 0.399 | 0.400 | 0.379 | 0.401 | 0.402 | 0.515 | 0.609 | 0.525 | 0.428 | 0.419 |
|                                                                        | taud        |        | 2.405 | 2.414 | 2.443 | 2.417 | 2.439 | 2.370 | 2.347 | 1.979 | 1.764 | 2.041 | 2.327 | 2.378 |
|                                                                        | Ebn at Noon |        | 295   | 291   | 287   | 272   | 265   | 250   | 254   | 233   | 225   | 257   | 289   | 294   |
|                                                                        | Edh at Noon |        | 40    | 39    | 36    | 35    | 31    | 32    | 34    | 53    | 71    | 56    | 43    | 42    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2120  | 1905  | 1765  | 1464  | 1086  | 941   | 1046  | 1276  | 1486  | 1721  | 1988  | 2100  |
|                                                                        | RadStd      |        | 144   | 146   | 109   | 135   | 157   | 103   | 118   | 88    | 150   | 161   | 125   | 128   |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page